



**NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY**

BACHELOR OF ARTIFICIAL INTELLIGENCE HONOURS

COMPUTATIONAL GAME THEORY

PAPER REVIEW

**LEARNING EQUILIBRIA IN COORDINATION GAMES VIA
MINORIZATION-MAXIMIZATION**

Krishnan, A., Le Cadre, H., & Busic, A. (2026). Learning equilibria in coordination games via minorization-maximization. arXiv preprint arXiv:2605.13644.

Submitted by: Muhupua Tjimune
Student number: 223075493

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1. Summary of the Paper

1.1 The Problem Being Addressed

In modern distributed data engineering and network design, such as managing smart electrical grids or organizing fleets of multi-agent autonomous robots, independent systems must coordinate to achieve a shared, system-wide objective while simultaneously managing their own internal operational costs. Standard game theory frameworks typically model these scenarios as "coordination games," operating under the strict assumption that all participating agents act with absolute mathematical rationality.

In real-world deployments, however, human participants or boundedly rational agents exhibit deep-seated psychological biases, such as an irrational aversion to risk or an unequal weighting of potential losses versus potential gains. This behavioral complexity introduces two major roadblocks for system design:

Equilibrium Multiplicity: The presence of human-like behavior creates an unpredictable landscape filled with multiple competing stable outcomes (equilibria), making it incredibly difficult to forecast or guarantee which final state the network will settle into.

Mathematical Non-Smoothness: Incorporating realistic behavioral models introduces mathematical irregularities, such as sharp bends, non-differentiable points, and localized traps in the objective functions. Because traditional optimization algorithms rely heavily on smooth calculus gradients, they easily stall, break down, or fail to compute when confronted with these rough behavioral surfaces.

1.2 The Main Contribution of the Work

To bridge the gap between behavioral psychology and decentralized optimization, the authors introduce a novel engineering framework. The core of their contribution is broken down into three primary elements:

A Realistic Behavioral Utility Model: Instead of using standard linear reward functions, the authors combine a collective "social purpose" team objective with individual actor payoffs that are explicitly transformed using Prospect Theory. This mathematically captures how real agents evaluate decisions based on a subjective reference point rather than absolute outcomes.

An Equilibrium Selection Mechanism via Regularization: To eliminate the chaos of multiple competing stable states, the authors introduce a mathematical smoothing technique known as *regularization* directly into the game's global potential function. This deliberate modification systematically smooths out the mathematical landscape, collapsing the field of multiple sub-optimal equilibria down into a single, predictable, globally optimal balance point.

A Decentralized Minorization-Maximization (MM) Learning Scheme: The authors design an iterative algorithm tailored for distributed networks. Rather than forcing independent agents to directly calculate

a highly complex, non-smooth optimization problem, the MM algorithm allows each agent to construct a simple, smooth, localized "surrogate" (substitute) function at every step. By maximizing this simpler substitute function iteratively, the agents collectively and reliably climb the mathematical landscape toward the ideal target behavior without needing global visibility.

1.3 The Theoretical and Simulation Results

The authors validate their framework through a combination of rigorous mathematical proofs and computer-based numerical simulations. Their findings deliver several key insights:

Guaranteed Global Convergence: The paper provides formal mathematical proofs showing that the proposed decentralized MM algorithm achieves a 100% success rate in converging directly to the single, optimal regularized equilibrium point, completely eliminating algorithmic stalling.

Superior Convergence Velocity: When tested in numerical simulations, the MM learning scheme demonstrated significantly faster convergence speeds compared to standard gradient-descent methods, making it highly attractive for environments requiring fast, iterative updates.

Robustness Against Local Traps: Traditional "best-response" algorithms frequently get stuck in local dead-ends (sub-optimal equilibria) because they only look one step ahead. The simulations proved that the surrogate-based approach of the MM algorithm successfully bypasses these mathematical traps, consistently reaching the global optimum across both smooth and highly irregular, non-smooth reward setups.

2. The Related Work

The authors position their research at a highly unique intersection, building directly upon three distinct bodies of established literature:

Coordination and Potential Games: The paper heavily relies on the foundations of potential game theory, where a single mathematical function can accurately reflect the alignment of player incentives across an entire system. The authors specifically expand on the sub-field of "social purpose games," pushing the literature forward by showing how community-level goals can coexist with decentralized, personalized operational costs.

Behavioral Game Theory and Economics: The framework integrates Nobel-prize-winning concepts from Prospect Theory, which was originally developed to explain why human economic actors fear financial losses far more intensely than they value equivalent gains. The authors bridge a massive gap by taking these purely observational, economic behavioral insights and turning them into actionable, algorithmic constraints for multi-agent engineering systems.

Algorithmic Optimization Methods: The Minorization-Maximization (MM) framework itself is a well-established mathematical tool traditionally used in centralized data science, machine learning, and image processing to solve difficult estimation problems. This paper expands on prior optimization literature by innovatively decentralizing the MM method, proving it can be broken apart and run step-by-step by independent, communicating actors across a network.

3. The Limitations of the Paper

While the framework represents a significant step forward, its practical application is bounded by three notable engineering limitations:

Dependence on an Intermediary Coordinator: Although the algorithm runs in a decentralized fashion among the agents, it still fundamentally requires a central, trusted intermediary coordinator to periodically aggregate network parameters and distribute information updates back to the participants. This architectural constraint prevents the algorithm from being deployed in purely ad-hoc, peer-to-peer networks that lack a permanent centralized infrastructure.

Static Assumption of Behavioral Risk Profiles: The model treats the agents' psychological risk preferences and behavioral parameters as fixed, permanent constants throughout the duration of the game. In actual real-world scenarios, human behavioral profiles are highly dynamic; individuals naturally adapt, shift their risk tolerance, and alter their decision-making strategies as they gain historical experience from interacting with the system.

High Upfront Analytical and Mathematical Overhead: The algorithm itself executes very quickly once running, but the preliminary phase requires a massive amount of manual, upfront mathematical engineering. Designing the specific regularized potential functions and calculating the tailored surrogate shapes requires deep domain expertise for every distinct application, which heavily limits its immediate "plug-and-play" usability for engineers deploying automated real-time systems.

References

Krishnan, A., Le Cadre, H., & Busic, A. (2026). Learning equilibria in coordination games via minorization-maximization. *arXiv preprint arXiv:2605.13644*.